Algebraic Anomalies in ECDSA Signatures: A Complete Mathematical Framework for Anomaly Classes A, B, C, D, and E

Identification Criteria, Recovery Formulas, Probabilistic Analysis, and Cross-Transaction Intersection Methods

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ABSTRACT

This document presents the complete mathematical framework for five classes of algebraic anomalies (A, B, C, D, E) identified in ECDSA signatures over elliptic curves. Each anomaly arises from structured arithmetic relationships within the standard signing equation $s = (z + rx)\,k^{-1} \bmod n$ under ideally random nonce generation. The anomalies divide into two fundamentally distinct categories. *Non-structural anomalies* (A, B, C, E) depend on exact arithmetic coincidences between internal signature components and occur with probability $O(n^{-1})$ or less per transaction; for three of these (A, B, E), closed-form deterministic private-key recovery formulas are derived and verified; class C admits identification but no known analytical recovery. *Structural anomaly D* is qualitatively different: it arises in approximately 25% of all transactions regardless of the curve order n, produces a bounded candidate search window for the hidden signing component, and enables private-key recovery via cross-transaction intersection of candidate sets on test curves. However, on production-scale curves such as secp256k1, typical window sizes exceed the tractability threshold by approximately 10^{38} orders of magnitude, rendering the naïve intersection method strictly inferior to generic discrete-logarithm attacks. The active research direction is the development of algebraic methods for exponential window reduction. A rigorous probabilistic analysis quantifies the expected frequency of each class, demonstrating that non-structural anomalies are inherent to the full transaction space but unobservable on production-scale curves such as secp256k1, while structural anomaly D remains universally present and is the subject of active research into computational tractability at scale.

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PART I

Foundations and Classification

3. Preliminaries

1.1 ECDSA Signing Equation

Let E be an elliptic curve defined over a finite field $\mathbb{F}_p$ with a base point G of prime order n. Given a private key $x \in \{1, \ldots, n-1\}$, a message hash $z \in \{1, \ldots, n-1\}$, and a random nonce $k \in \{1, \ldots, n-1\}$, the ECDSA signature (r, s) is computed as:

$$r = \left[k \cdot G\right]_x \bmod n, \qquad s = (z + r\,x)\,k^{-1} \bmod n \tag{1}$$

where $[\cdot]_x$ denotes the x-coordinate of the resulting elliptic curve point.

1.2 Component Decomposition

The signature scalar s admits an additive decomposition into two components that separate the message-hash contribution from the private-key contribution:

$$s_{zk} = z \cdot k^{-1} \bmod n, \qquad s_{rxk} = r \cdot x \cdot k^{-1} \bmod n \tag{2}$$

$$s = (s_{zk} + s_{rxk}) \bmod n \tag{3}$$

Both s_{zk} and s_{rxk} are internal (hidden) quantities that cannot be directly observed from the public signature (r, s) alone. The anomaly classification framework identifies structural conditions under which algebraic relationships between publicly derivable quantities constrain s_{zk} to a recoverable value or a bounded set of candidates.

1.3 Auxiliary Parameters

> ***Definition 1 (Auxiliary modular product).***
>
> Define the auxiliary quantity $s_{zr} = z \cdot r \bmod n$, fully computable from the public signature components.
>
> $$s_{zr} = z \cdot r \bmod n \tag{4}$$

> ***Definition 2 (Anomaly residue).***
>
> For a transaction satisfying $s_{zr} > s$, define $d = (s_{zr} - s) \bmod n$ and the residue:
>
> $$a = s \bmod d \tag{5}$$
>
> A transaction qualifies for anomaly classification (types A, B, C, D) if and only if $a > 0$, $a \neq s$, and $a \neq s_{zr}$.

> ***Definition 3 (Divisibility ratios).***
>
> For a qualified transaction, define two rational-valued diagnostic ratios:
>
> $$m_1 = \frac{s_{zk}}{a}, \qquad m_2 = \frac{s + s_{zr}}{s_{rxk}} \tag{6}$$
>
> where the divisions are performed over $\mathbb{Q}$ (exact rational fractions), not modular arithmetic.

4. Classification Framework

2.1 Residue-Based Classification (A–D)

Class	Condition on m_1	Condition on m_2	Additional
A	$m_1 \in \mathbb{Z}, m_1 = 1$	$m_2 \notin \mathbb{Z}$	—
B	$m_1 \in \mathbb{Z}$	$m_2 \in \mathbb{Z}$	$m_1 = m_2$

C	$m_1 \in \mathbb{Z}, m_2 \in \mathbb{Z}, m_1 \neq m_2$; *or* $m_1 \in \mathbb{Z}, m_1 > 1, m_2 \notin \mathbb{Z}$		—
D	$m_1 \notin \mathbb{Z}$	(any)	—

Table 1. Decision rules for anomaly classification based on the divisibility ratios m_1 and m_2.

2.2 Independent Classification (E)

Anomaly class E is evaluated independently of the residue-based classification, prior to the A–D decision tree, via a symmetry condition on the additive decomposition $s = s_{zk} + s_{rxk}$ (Section 6).

2.3 Taxonomic Dichotomy

Non-structural anomalies (A, B, C, E): depend on exact integer divisibility or precise additive symmetry. Probability is $O(n^{-1})$ or smaller per transaction and vanishes for large n. Recovery (where possible) is deterministic from a single transaction.

Structural anomaly (D): the generic, overwhelmingly dominant case where $m_1 \notin \mathbb{Z}$. It occurs at a fixed fraction $\approx 1/4$ of all transactions, independent of n. Recovery requires enumeration over a bounded candidate window and intersection across multiple transactions.

PART II

Non-Structural Anomalies: Classes A, B, C, and E

5. Anomaly Class A: Linear Divisibility with Unit Multiplier

3.1 Identification Criterion

> ***Definition 4 (Anomaly A).***
>
> A qualified transaction exhibits anomaly A if and only if:
>
> $$m_1 = \frac{s_{zk}}{a} = 1, \qquad m_2 \notin \mathbb{Z} \tag{7}$$
>
> Equivalently, $s_{zk} = a$ as integers.

This condition asserts that the hidden component s_{zk} is exactly equal to the publicly computable residue a, immediately revealing its value.

3.2 Private-Key Recovery Formula

Theorem 1 (Closed-form recovery for Anomaly A).

Given a transaction exhibiting anomaly A with public parameters (s, z, r, n) and residue a, the private key x is recovered deterministically:

Step 1. Set $s_{zk} = a$.
Step 2. Recover the nonce: $k = s_{zk} \cdot z^{-1} \bmod n$ *(8)*
Step 3. Compute: $s_{rxk} = (s - s_{zk}) \bmod n$ *(9)*
Step 4. Recover the private key:

$$x = s_{rxk} \cdot (r \cdot k)^{-1} \bmod n \tag{10}$$

Remark 1.

All inversions exist with probability $1 - O(1/n)$ since n is prime and $z, r, k \not\equiv 0 \pmod{n}$ by the ECDSA specification.

6. Anomaly Class B: Dual Integral Divisibility

4.1 Identification Criterion

Definition 5 (Anomaly B).

A qualified transaction exhibits anomaly B if and only if:

$$m_1 \in \mathbb{Z}, \qquad m_2 \in \mathbb{Z}, \qquad m_1 = m_2 \tag{11}$$

This is the rarest of all five classes, requiring simultaneous exact divisibility of two independent modular quantities with coinciding quotients.

4.2 Derivation of the Quadratic Recovery Equation

Let $m = m_1 = m_2$. From $s_{zk} = am$ and $s + s_{zr} = s_{rxk} \cdot m = (s - am)m$, rearranging:

$$am^2 - sm + (s + s_{zr}) \equiv 0 \pmod{n} \tag{12}$$

4.3 Private-Key Recovery Formula

Theorem 2 (Quadratic recovery for Anomaly B).

Step 1. Compute the discriminant:

$$D = \left(s^2 - 4\,a\,(s + s_{zr})\right) \bmod n \tag{13}$$

Step 2. Compute modular square roots $\sqrt{D} \bmod n$.

Step 3. For each root ρ, compute the candidate multiplier:

$$m = (s + \rho) \cdot (2a)^{-1} \bmod n \tag{14}$$

Step 4. Recover the private key:

$$s_{zk} = a \cdot m \bmod n, \quad k = s_{zk} \cdot z^{-1} \bmod n, \quad x = (s - s_{zk}) \cdot (r \cdot k)^{-1} \bmod n \tag{15}$$

Remark 2.

The modular square root exists if and only if D is a quadratic residue modulo n (testable via the Euler criterion). The quadratic equation generically produces two candidates for m; both must be tested.

7. Anomaly Class C: Partial Divisibility

5.1 Identification Criterion

Definition 6 (Anomaly C).

A qualified transaction exhibits anomaly C if:

(i) $m_1 \in \mathbb{Z}$, $m_2 \in \mathbb{Z}$, but $m_1 \neq m_2$; *or*
(ii) $m_1 \in \mathbb{Z}$ with $m_1 > 1$, and $m_2 \notin \mathbb{Z}$.

The residue a divides s_{zk} exactly, but the multiplier m_1 is a free integer parameter that cannot be determined from public data alone:

$$s_{zk} = a \cdot m_1, \qquad m_1 \in \{2, 3, 4, \ldots\} \tag{16}$$

5.2 Status of Analytical Recovery

> ***Remark 3.***
>
> No closed-form analytical formula for deterministic private-key recovery has been derived for anomaly class C. The residue a constrains s_{zk} to a lattice $\{a, 2a, 3a, \ldots\} \cap [1, n-1]$, yielding approximately $\lfloor (n-1)/a \rfloor - 1$ candidate values—computationally infeasible for secp256k1 without further algebraic reduction.

8. Anomaly Class E: Goldbach-Type Additive Decomposition

6.1 Identification Criterion

Anomaly E is evaluated independently of the residue-based classification and operates directly on the additive decomposition $s = s_{zk} + s_{rxk}$, possibly extended modulo n.

> ***Definition 7 (Anomaly E1).***
>
> A transaction exhibits anomaly E1 if: (i) $s_{zk} + s_{rxk} = s$ (no wrap-around); (ii) s is even; (iii) $|s_{zk} - s_{rxk}| = \lfloor s/2 \rfloor + 1$.

> ***Definition 8 (Anomaly E2).***
>
> A transaction exhibits anomaly E2 if: (i) $s_{zk} + s_{rxk} > s$, so $\tilde{s} = s + n$; (ii) $\tilde{s}$ is even; (iii) $|s_{zk} - s_{rxk}| = \lfloor \tilde{s}/2 \rfloor + 1$.

The condition constrains s_{zk} to at most two specific values from $[1, n-1]$.

6.2 Private-Key Recovery Formula

> ***Theorem 3 (Goldbach-type recovery for Anomaly E).***
>
> Given effective sum σ ($= s$ for E1 or $s + n$ for E2):
>
> **Step 1.** Compute two candidate decomposition values:
>
> $$\alpha = \left\lfloor \frac{\sigma + \lfloor \sigma/2 \rfloor + 1}{2} \right\rfloor, \qquad \beta = \sigma - \alpha \tag{17}$$
>
> **Step 2.** For each candidate $c \in \{\alpha, \beta\}$, recover:
>
> $$k_c = c \cdot z^{-1} \bmod n, \qquad x_c = (s - c) \cdot (r \cdot k_c)^{-1} \bmod n \tag{18}$$

Step 3. The correct private key is one of $\{x_\alpha, x_\beta\}$, disambiguated by verification against the public key.

Remark 4.

The recovery produces exactly two candidates. A single scalar multiplication $x_c \cdot G \stackrel{?}{=} Q$ suffices to identify the correct value.

PART III

Structural Anomaly D

9. Anomaly Class D: Structural Factor-Level Decomposition

7.1 Identification Criterion

Definition 9 (Anomaly D).

A qualified transaction exhibits anomaly D if and only if:

$$f = \frac{s_{zk}}{a} \notin \mathbb{Z} \tag{19}$$

Equivalently, $a \nmid s_{zk}$ in ordinary integer arithmetic.

Definition 10 (Level index).

The *level* of an anomaly-D transaction is:

$$N = \lfloor f \rfloor = \left\lfloor \frac{s_{zk}}{a} \right\rfloor, \qquad 0 \leq N \leq \left\lfloor \frac{n-1}{a} \right\rfloor \tag{20}$$

A transaction at level N is denoted $\mathrm{D}N$ (e.g., $\mathrm{D0}$, $\mathrm{D14}$, $\mathrm{D111}$).

7.2 The Factor Window

Theorem 4 (Factor window).

If $\lfloor s_{zk}/a \rfloor = N$, then $s_{zk} \in [aN,\ a(N+1))$.

Proof. From $N \leq s_{zk}/a < N+1$ it follows that $aN \leq s_{zk} < a(N+1)$. $\square$

Definition 11 (Search range).

$$\mathcal{R}(a, N) = \begin{cases} \{1, 2, \ldots, a\} & \text{if } N = 0 \\ \{aN,\ aN+1,\ \ldots,\ a(N+1)\} & \text{if } N \geq 1 \end{cases} \tag{21}$$

with $|\mathcal{R}(a, N)| = a$ for $N = 0$ and $a+1$ for $N \geq 1$.

7.3 Candidate Private-Key Recovery

Theorem 5 (Candidate recovery formula for Anomaly D).

For each candidate $\hat{s}_{zk} \in \mathcal{R}(a, N)$, the candidate private key is:

$$\hat{x} = z \cdot (s - \hat{s}_{zk}) \cdot (r \cdot \hat{s}_{zk})^{-1} \bmod n \tag{22}$$

Proof. From $s_{rxk} = (s - s_{zk}) \bmod n$ and $k = z \cdot s_{zk}^{-1} \bmod n$:
$x = (s - s_{zk})(r \cdot s_{zk} \cdot z^{-1})^{-1} = z(s - s_{zk})(r \cdot s_{zk})^{-1} \bmod n$. $\square$

Remark 5.

Equation (22) defines a bijection over $\{1, \ldots, n-1\}$: the correct private key x appears exactly once at $\hat{s}_{zk} = s_{zk}$.

10. Cross-Transaction Intersection Recovery for Anomaly D

8.1 Foundational Principle

A single anomaly-D transaction produces $\approx a+1$ candidate private keys. If multiple transactions share the same private key x, the correct x must appear in every candidate set.

Definition 12 (Candidate set).

$$\mathcal{X}_i(N_i) = \left\{ z_i(s_i - \hat{s}_{zk})(r_i \hat{s}_{zk})^{-1} \bmod n \;\middle|\; \hat{s}_{zk} \in \mathcal{R}(a_i, N_i) \right\} \tag{23}$$

Theorem 6 (Intersection principle).

For T transactions signed with the same key x, with correct levels $N_1, \ldots, N_T$:

$$x \in \bigcap_{i=1}^{T} \mathcal{X}_i(N_i) \tag{24}$$

The expected intersection cardinality (under the uniform heuristic for false positives):

$$\mathbb{E}\left[\left|\bigcap_{i=1}^{T} \mathcal{X}_i\right|\right] \approx 1 + \frac{\prod_{i=1}^{T} w_i}{(n-1)^{T-1}} \tag{25}$$

where $w_i = |\mathcal{R}(a_i, N_i)|$.

Corollary 1 (Uniqueness condition).

The intersection yields a unique result when:

$$\prod_{i=1}^{T} w_i < (n-1)^{T-1} \tag{26}$$

8.2 Computational Gap Analysis on secp256k1

The uniqueness condition (26) imposes a hard constraint on the window sizes w_i. For T transactions with uniform window size W, uniqueness requires $W^T < (n-1)^{T-1}$, equivalently:

$$W < (n-1)^{(T-1)/T} \tag{27}$$

For secp256k1 ($n \approx 1.158 \times 10^{77}$) with $T = 2$ transactions, this yields $W < \sqrt{n} \approx 10^{38.5}$.

However, the window size $W = |\mathcal{R}(a, N)| \approx a + 1$ is determined by the anomaly residue $a = s \bmod d$, where $d = (s_{zr} - s) \bmod n$. Since s and d are pseudorandom elements of $\mathbb{Z}_n$, the residue a is distributed over $[1, d-1]$ with typical values of order $O(d)$. For uniformly distributed d, the expected value of a is:

$$\mathbb{E}[a] \approx \frac{n}{4} \approx 2.9 \times 10^{76} \tag{28}$$

Proposition 2 (Fundamental computational barrier).

On secp256k1, the typical window size $W \approx n/4 \approx 2.9 \times 10^{76}$ exceeds the uniqueness threshold $\sqrt{n} \approx 10^{38.5}$ by approximately **38 orders of magnitude**. Consequently:

(i) Even the additive intersection cost $O(\sum w_i)$ with typical window sizes is $O(n)$ per transaction—equivalent to exhaustive search over the full key space.

(ii) The best known generic attack on the elliptic curve discrete logarithm problem (Pollard's rho / Baby-step Giant-step) requires $O(\sqrt{n}) \approx 10^{38.5}$ group operations. The naïve anomaly-D

> intersection with typical window sizes requires $O(n) \approx 10^{77}$ modular operations, which is $\sim 10^{38}$ **times worse** than the standard ECDLP attack that is itself already considered computationally infeasible.
>
> *(iii)* The probability of encountering a transaction with $a < \sqrt{n}$ (which would bring the window below the uniqueness threshold for $T = 2$) is approximately $\sqrt{n}/n = 1/\sqrt{n} \approx 10^{-38.5}$. Observing even one such transaction from a given key requires on the order of $10^{38.5}$ same-key transactions, which is physically unrealizable.

> ***Remark 6 (Honest assessment of current state).***
>
> The intersection framework as described above is *mathematically correct* and *experimentally validated* on test curves where n is small enough that $a + 1 \ll n$. On test_small ($n \approx 10^5$), typical $a \approx 10^4$ and $\sqrt{n} \approx 316$; since $a > \sqrt{n}$, even on test curves the method does not satisfy the uniqueness condition for $T = 2$ with generic transactions—it succeeds because the test demonstrates correct mechanics with *known* levels, not because the window is inherently narrow.
>
> On secp256k1, the current intersection method without window reduction is not a viable attack vector. It is strictly inferior to generic ECDLP algorithms by a factor of $\sqrt{n}$. The research direction is therefore not the intersection algorithm itself, but the development of **algebraic methods to exponentially narrow the s_{zk} candidate window** below the $O(a)$ bound—a problem that remains open.

8.3 Intersection Algorithm

Algorithm 1: Cross-Transaction Intersection for Anomaly D

Input: Transactions $\{T_1, \ldots, T_T\}$ with public parameters (s_i, z_i, r_i, a_i) and curve order n.

Output: Set of candidate private keys $\mathcal{X}^*$.

1. For each T_i, compute $\mathcal{R}(a_i, N_i)$ for the assumed level N_i.
2. Sort transactions by ascending window size.
3. Build the *base index* from the narrowest transaction: for each $\hat{s}_{zk} \in \mathcal{R}(a_{\pi(1)}, N_{\pi(1)})$, compute $\hat{x}$ via (22) and store in a hash map $\mathcal{M}$.
4. For each subsequent $T_{\pi(j)}$: build local index, intersect with $\mathcal{M}$.
5. Return $\mathcal{X}^* = \mathbf{keys}(\mathcal{M})$.

Remark 7 (Complexity).

Total cost is $O\left(\sum_{j=1}^{T} w_j \cdot \log^2 n\right)$—additive rather than the multiplicative $\prod w_i$ of the naïve Cartesian product.

11. Level Distribution of Anomaly D

9.1 Hyperbolic Decay Law

Theorem 7 (Marginal level distribution).

The probability that an anomaly-D transaction has level N follows an approximate hyperbolic law:

$$\Pr[N] \propto \frac{1}{N+c} \tag{29}$$

for a constant $c > 0$. As N increases, fewer transactions have sufficiently small a to reach that level (since $aN < n$ is required), producing the observed decay.

9.2 Empirical Level Distributions

Level N	test_small ($n \approx 10^5$)	test_large ($n \approx 10^9$)	Ratio to D0
0	17,567	17,529	1.000
1	17,632	17,456	≈ 1.00
2	17,189	17,542	≈ 0.98
3	16,743	16,925	≈ 0.96
5	12,860	13,021	≈ 0.73
10	6,301	6,193	≈ 0.36
20	2,448	2,451	≈ 0.14
50	620	573	≈ 0.035
100	199	194	≈ 0.011

Table 2. Selected level counts for anomaly-D transactions (10^6-transaction samples).

Remark 8 (Scale invariance).

The distributions are remarkably consistent across curves differing by four orders of magnitude in n. Levels D0–D3 are approximately equiprobable ($\approx 17{,}000$ each), with smooth hyperbolic decay thereafter. This confirms that the level structure is a *scale-invariant* property of ECDSA modular arithmetic.

PART IV

Probabilistic Analysis and Empirical Validation

12. Probabilistic Analysis

10.1 Transaction Space

The total number of distinct valid transactions is $|\mathcal{T}| = (n-1)^4$, corresponding to the four free parameters (z, x, k, r) with s determined by the signing equation (1).

10.2 Non-Structural Anomalies

Anomaly A: requires $s_{zk} = a$ (one specific value out of $n-1$):

$$\Pr[\mathrm{A}] \approx \frac{1}{4(n-1)}, \qquad \mathbb{E}[\#A] = \frac{N_{\mathrm{tx}}}{4(n-1)} \tag{30}$$

Anomaly B: requires dual divisibility with coinciding quotients:

$$\Pr[\mathrm{B}] \sim O(n^{-2}), \qquad \mathbb{E}[\#B] \approx \frac{N_{\mathrm{tx}}}{4n^2} \tag{31}$$

Anomaly C: requires $a \mid s_{zk}$ with $m_1 \geq 2$; averaging over the heavy-tailed distribution of a:

$$\mathbb{E}[\#C] \approx \frac{N_{\mathrm{tx}}}{4} \cdot \frac{c \cdot \ln n}{n} \tag{32}$$

where $c > 1$ is a bias correction factor.

Anomaly E: fixes s_{zk} to at most 2 values; exactly one of $s, s+n$ is even for prime $n > 2$:

$$\Pr[\mathrm{E}] = \frac{1}{n-1}, \qquad \mathbb{E}[\#E] = \frac{N_{\mathrm{tx}}}{n-1} \tag{33}$$

10.3 Structural Anomaly D

$$\Pr[\mathrm{D}] \approx \Pr[\text{qualified}] \cdot \Pr[a \nmid s_{zk}] \approx \frac{1}{4}\left(1 - O\left(\frac{\ln n}{n}\right)\right) \approx \frac{1}{4} \tag{34}$$

This fraction is independent of n.

10.4 Summary

Anomaly	Per-transaction probability	Order	Nature
A	$\frac{1}{4(n-1)}$	$O(n^{-1})$	Non-structural
B	$\frac{1}{4n^2}$	$O(n^{-2})$	Non-structural
C	$\frac{c \ln n}{4n}$	$O(n^{-1} \ln n)$	Non-structural
E	$\frac{1}{n-1}$	$O(n^{-1})$	Non-structural
D	$\approx 1/4$	$O(1)$	**Structural**

Table 3. Asymptotic per-transaction probability of each anomaly class.

13. Numerical Validation

11.1 Test Curves

Curve	n	N_{tx}	Qualified	#A	#B	#C	#D	#E
test_small	99,667	10^6	249,410	4	0	580	248,814	12
test_large	1.24×10^9	10^6	250,311	0	0	0	250,311	0

Table 4. Observed anomaly counts across test curves.

> ***Remark 9.***
> On test_small: $\mathbb{E}[\#A] \approx 2.5$ (observed 4), $\mathbb{E}[\#E] \approx 10$ (observed 12)—within Poisson fluctuation. Anomaly C observed count (580) exceeds the naïve estimate by $\sim 10\times$, explained by the heavy-tailed distribution of the residue a. On test_large ($n \approx 10^9$), no non-structural anomalies are observed ($\mathbb{E}[\#A] \approx 2 \times 10^{-4}$). Anomaly D dominates both datasets at $\approx 25\%$.

11.2 Predictions for secp256k1

Curve order: $n \approx 1.158 \times 10^{77}$. Sample: $N_{\text{tx}} = 10^{12}$.

Anomaly	$\mathbb{E}[\#]$ at $N_{\text{tx}} = 10^{12}$	Observability
A	$\approx 2.2 \times 10^{-66}$	Impossible
B	$\approx 1.9 \times 10^{-143}$	Impossible
C	$\approx 3.8 \times 10^{-64}$	Impossible
E	$\approx 8.6 \times 10^{-66}$	Impossible
D	$\approx 2.5 \times 10^{11}$	**Always present**

Table 5. Expected anomaly counts for secp256k1.

To observe a single non-structural anomaly with probability $> 1/2$:

$$N_{\text{min}} \approx \frac{n \ln 2}{4} \approx 2.0 \times 10^{76} \text{ transactions} \tag{35}$$

This exceeds any physically realizable computation.

11.3 Structural Existence

> ***Proposition 1.***
>
> For any elliptic curve with prime order $n \geq 5$, the full space $\mathcal{T} = \{1, \ldots, n-1\}^4$ contains:
>
> $$|\mathcal{A}_{\text{A}}| \sim \frac{(n-1)^3}{4}, \quad |\mathcal{A}_{\text{C}}| \sim \frac{c(n-1)^3 \ln n}{4}, \quad |\mathcal{A}_{\text{E}}| \sim (n-1)^3 \tag{36}$$
>
> These are astronomically large in absolute terms ($\sim 10^{231}$ for secp256k1), yet constitute a vanishing $O(n^{-1})$ fraction of $|\mathcal{T}| \sim 10^{308}$.

11.4 Intersection Example

Four transactions with $x = 32{,}768$ on test_small ($n = 99{,}667$):

Tx	a_i	f_i	N_i	$\vert\mathcal{R}\vert$
T_1	3,281	11.254	11	3,282
T_2	5,725	0.741	0	5,725
T_3	4,376	4.762	4	4,377
T_4	12,242	0.687	0	12,242

Table 6. Example transactions for intersection recovery.

Cartesian product: $\approx 1.01 \times 10^{15}$. Algorithm 1 cost: $25{,}626$ operations. Result: unique $x = 32{,}768$ recovered in < 0.1 s.

14. Comparative Summary and Open Problems

12.1 Comparative Table

Property	Anomalies A, B, C, E	Anomaly D
Nature	Exact arithmetic coincidence	Structural; generic case
Per-tx probability	$O(n^{-1})$ to $O(n^{-2})$	$\approx 1/4$ (constant)
Dependence on n	Vanishes as $n \to \infty$	Independent of n
Recovery type	Closed-form (A,B,E) or none (C)	Candidate-set + intersection
Single-tx recovery	Deterministic (A,B,E)	Requires enumeration over $\mathcal{R}$
Multi-tx needed	No (A,B,E)	Essential
Observability (secp256k1)	≈ 0	$\approx 25\%$ of all tx
Naïve cost on secp256k1	N/A (unobservable)	$O(n) \approx 10^{77}$ ops (typical a)
vs. generic ECDLP attack	N/A	$\sim 10^{38}\times$ worse than BSGS ($O(\sqrt{n})$)

Table 7. Comparative summary of all anomaly classes.

12.2 Conclusions

Anomalies A, B, and E provide instant deterministic key recovery but occur with vanishing probability on production curves—mathematically elegant but cryptographically irrelevant for secp256k1. Their security significance lies not in their absence (they exist in $\sim 10^{231}$ instances), but in the impossibility of encountering them in any feasible sample.

Anomaly D is structurally distinct: it affects one in four transactions on *any* curve and provides a bounded candidate search space characterized by the factor-level decomposition. The intersection algorithm reduces the combinatorial explosion from a multiplicative Cartesian product to additive complexity, which enables efficient recovery on test curves. However, this efficiency gain does not overcome the fundamental barrier on secp256k1: with typical window sizes $W \approx n/4$, even the additive cost is $O(n)$—approximately 10^{38} times worse than the best generic ECDLP attacks (which are themselves considered infeasible).

The research program is therefore not directed at applying the current intersection method to secp256k1 (which would be futile), but at developing **algebraic methods for exponential window reduction**—techniques that would narrow the s_{zk} candidate range from $O(a)$ to a computationally tractable size without relying on the statistical rarity of small a. Whether such methods exist is the central open question.

12.3 Open Problems

> ***Open Problem 1 (Exponential window reduction).***
>
> The central barrier to anomaly-D exploitation on secp256k1 is the window size: typical $a \approx n/4 \approx 10^{76}$, while tractability requires $W \ll \sqrt{n} \approx 10^{38.5}$. The gap is $\sim 10^{38}$ orders of magnitude. Bridging this gap requires discovering algebraic structure in the relationship between a, s_{zk}, and s that enables exponential reduction of the candidate range—i.e., narrowing $\mathcal{R}(a, N)$ from $O(a)$ candidates to $O(\mathrm{poly}(\log n))$ without knowledge of the true level N. No such method is currently known. Without it, anomaly-D intersection is strictly inferior to generic ECDLP algorithms on secp256k1 by a factor of $\sqrt{n} \approx 10^{38.5}$.

> ***Open Problem 2 (Recovery for Anomaly C).***
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> Deriving a closed-form formula or algebraic reduction that avoids exhaustive enumeration over the lattice $\{a, 2a, 3a, \ldots\}$.

> ***Open Problem 3 (Residue distribution).***
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> The discrepancy between the naïve C-estimate ($c \approx 2$) and observed count ($c_{\text{eff}} \approx 20$) indicates that $a = s \bmod d$ has heavier tails than predicted. A precise analytical characterization would sharpen all frequency predictions and inform the window-size distribution for anomaly-D attacks.

Notation. All arithmetic is modulo n unless stated otherwise. a^{-1}: modular inverse of a mod n. $\lfloor \cdot \rfloor$: floor function. $\mathcal{R}(a, N)$: candidate search range for residue a at level N. $|\cdot|$: set cardinality. N_{tx}: total sampled transactions.

Status. Anomalies A, B, E: verified closed-form recovery. Anomaly C: no known analytical recovery. Anomaly D: under active research. Intersection framework validated on test curves ($n \approx 10^5, 10^9$). On secp256k1 ($n \approx 10^{77}$), typical window sizes exceed tractability thresholds by $\sim 10^{38}$ orders of magnitude; algebraic methods for exponential window reduction are being investigated.